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System and methodology for bypassing through an expandable metal packer

Abstract

A technique facilitates improved actuation and use of packers which may be disposed along a well string placed in a borehole, e.g. a wellbore. Each packer may be constructed with a tubing, sometimes referred to as a mandrel, and an expandable metal bladder secured around the tubing via connections. The packer may further comprise a sealing element mounted about the expandable metal bladder. The packer also comprises couplings, e.g. a first coupling and a second coupling, connected at opposite ends of the tubing. The couplings each having a hole therethrough to provide a feedthrough running from an exterior to an interior of the tubing at both ends of the tubing.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a national stage entry under 35 U.S.C. 371 of International Application No. PCT/US2022/047653, filed Oct. 25, 2022 which claims the benefit of U.S. Provisional Application No. 63/273,493 entitled “System

and Methodology for Bypassing Through an Expandable Metal Packer,” filed Oct. 29, 2021, the disclosure of which is incorporated by reference in its entirety.

BACKGROUND

(1) In many oil and gas well applications, a wellbore is drilled into the earth and through a reservoir of a desired fluid, e.g. oil and/or gas. The wellbore may subsequently be completed with appropriate completion equipment having packers which may be expanded to isolate regions along the wellbore. For example, packers may be disposed along sand control equipment or other types of completion equipment to facilitate production of the desired fluids from the reservoir. Depending on the application, the packers may be mounted along a well string and selectively expanded to effectively form a seal between production tubing of the well string and the surrounding wellbore wall. In some applications, the completion equipment may comprise alternate path systems, control lines, and/or other components which extend down along the production tubing. However, routing such components through existing packer designs while maintaining desired packer functionality can be expensive and/or problematic.

SUMMARY

(2) In general, a system and methodology facilitate improved actuation and use of packers disposed along a well string and placed in a borehole, e.g. a wellbore. Each packer may comprise a tubing, sometimes referred to as a mandrel, and an expandable metal bladder secured around the tubing via connections. The packer may further comprise a sealing element mounted about the expandable metal bladder. The packer further comprises couplings, e.g. a first coupling and a second coupling, connected at opposite ends of the tubing. The couplings each have at least one hole therethrough to provide a feedthrough running from an exterior to an interior of the tubing. This allows a component to be fed from an exterior of the packer through the hole of one coupling; along an interior of the tubing; and out to the exterior of the packer through the hole of the other coupling. The component may comprise various types of control lines, e.g. hydraulic, electric, optical fiber; alternate path tubes; or other components which bypass the packer.

(3) However, many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein, and:

(2) FIG. 1 is an illustration of an example of a packer utilizing an expandable metal bladder and at least one feedthrough, according to an embodiment of the disclosure;

(3) FIG. 2 is a cross-sectional illustration of the packer shown in FIG. 1 without a bypass component, according to an embodiment of the disclosure;

(4) FIG. 3 is a cross-sectional illustration of the packer shown in FIG. 1 with a bypass component extending therethrough, according to an embodiment of the disclosure; and

(5) FIG. 4 is a cross-sectional illustration of another example of a packer having at least one feedthrough, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

(6) In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that the system and/or methodology may be practiced without these details and that

numerous variations or modifications from the described embodiments may be possible.

(7) The disclosure herein generally involves a system and methodology which facilitate improved actuation and use of packers disposed in a wellbore or other type of borehole. According to an embodiment, the packer may be constructed with a tubing, sometimes referred to as a mandrel, and an expandable metal bladder secured around the tubing via connections. The packer may further comprise a sealing element mounted about the expandable metal bladder. The packer further comprises couplings, e.g. a first coupling and a second coupling, connected at opposite ends of the tubing. The couplings each have at least one hole therethrough to provide a feedthrough running from an exterior to an interior of the tubing at both ends of the tubing. This allows a component to be fed from an exterior of the packer through the hole of one coupling; along an interior of the tubing; and out to the exterior of the packer through the hole of the other coupling. The component may comprise various types of control lines, e.g. hydraulic, electric, optical fiber; alternate path tubes; or other components which bypass the packer.

(8) An individual packer or a plurality of packers may be disposed along a well string to enable isolation of zones/regions along the wellbore or other type of borehole. For example, a plurality of the packers may be disposed along a well string having a downhole completion comprising sand control equipment. The packers enable isolation of well zones from which a production fluid, e.g. oil and/or gas, is received and produced to a desired collection location. Once positioned downhole, the packer or packers may be actuated by expanding the expandable metal bladder to drive the sealing element into sealing engagement with the surrounding borehole wall. The expandable metal bladder may be expanded and plastically deformed via application of sufficient pressure along its interior. For example, the expandable metal bladder may be expanded via hydroforming by directing fluid under pressure down through an interior of the well string and to an interior of the expandable metal bladder.

(9) The packer described herein provides an innovative arrangement for placing feedthrough lines or other feedthrough components through the packer. Effectively, the structure enables a bypass through the length of the packer so the lines or other components are able to easily bypass the packer without affecting operation of the packer.

(10) Referring generally to FIG. 1, an example of a packer **20** is illustrated as positioned along a well string **22** within a borehole **24**, e.g. a wellbore. In this example, the packer **20** is in the form of an expandable metal packer having a tubing **26**, sometimes referred to as a mandrel. An expandable metal bladder **28** is secured along the tubing **26** and secured around the tubing **26** via suitable connections **30**. The connections **30** may be in the form of end connections, as illustrated, coupled to opposite ends of the expandable metal bladder **28**.

(11) The connections **30** may be sealingly secured to the expandable metal bladder **28** and to the tubing **26** via welding, crimping, using seals and locking mechanisms, or via other suitable connection techniques. The sealed engagement effectively forms a packer chamber **32** which is an internal cavity between the expandable metal bladder **28** and the tubing **26**, as further illustrated in FIGS. 2 and 3. The expandable metal bladder **28** may be selectively expanded via pressurized fluid delivered down through an internal passage **34** of the well string **22**/tubing **26** and directed to packer chamber **32** via a suitable lateral expansion port or ports **36** extending laterally through tubing **26** as illustrated. In some embodiments, flow through port(s) **36** may be controlled by a valve or other flow control device.

(12) A sealing element **38**, e.g. an elastomeric sealing element, may be adhered or otherwise secured about the expandable metal bladder **28**. The sealing element **38** is positioned to move into sealing engagement with a surrounding borehole wall **40** defining borehole **24**. When the expandable metal bladder **28** is sufficiently expanded via application of suitable internal pressure, the expandable metal bladder **28** is caused to expand and plastically deform so as to securely force the sealing element **38** against borehole wall **40**.

(13) With additional reference to FIGS. 1-3, the expandable metal packer **20** also comprises

couplings **42** connected to the tubing **26**. For example, the couplings **42** may comprise first and second couplings connected to first and second (opposite) ends of the tubing **26**. The couplings **42** may be connected to tubing **26** via suitable threaded connections, welding, crimping, using seals and locking mechanisms, or via other suitable connection techniques. In some embodiments, the tubing **26** and the first and second couplings **42** may be formed as a unitary, single part.

(14) The couplings **42** also may be connected with corresponding uphole and downhole tubing sections **44** which form part of the overall well string **22**. In this manner, the packer **20** may be coupled into well string **22** via threaded connection or other suitable connection between the corresponding couplings **42** and tubing sections **44**. It should be noted that in some embodiments, the tubing **26** and the tubing sections **44** may have the same diameter, e.g. the same interior diameter.

(15) Additionally, each coupling **42** comprises a hole **46** or a plurality of holes **46** arranged to help provide a packer bypass for a well component **48**, e.g. a control line or other bypass line **50**, as illustrated in FIG. 3. Each control line **50** may comprise a hydraulic control line, an electric control line, an optical fiber control line, or another type of control line routed so as to bypass packer **20**.

(16) In some embodiments, a plurality of control lines **50** may be routed to bypass packer **20**. The plurality of control lines **50** may comprise similar or dissimilar control lines. For example, the control lines **50** bypassing packer **20** may include various combinations of hydraulic, electric, and/or optical fiber control lines. Additionally, a plurality of similar or dissimilar components **48**/control lines **50** may be routed through separate dedicated holes **46** in each coupling **42**. In other embodiments, however, groups of two or more components **48**/control lines **50** may be routed collectively through sufficiently large individual holes **46** in each coupling **42**.

(17) The hole or holes **46** in each coupling **42** may be drilled or otherwise formed to create a passage between an exterior **52** of the packer **20**/tubing **26** and the internal passage **34** of tubing **26**. This allows the component(s) **48**, e.g. control line(s) **50**, to bypass the packer **20** along an interior of the tubing **26** between the two couplings **42**. The control lines **50** (or other components) may be sealed to each coupling **42** at the corresponding hole or holes **46** via a sealing connector **54**, e.g. a dry mate connector or other suitable sealing connector.

(18) In some embodiments, a protection tube **56** may be placed inside the tubing **26** and connected to an interior region of the couplings **42**. The protection tube **56** may be connected at its opposite ends to couplings **42** via a locking element **58** which may be in the form of an abutment, lock ring, weldment, or other suitable element to hold protection tube **56** to both couplings **42** within the surrounding tubing **26**.

(19) The protection tube **56** engages the couplings **42** on opposite side of the hole(s) **46** relative to tubing **26**. This arrangement effectively creates a protected passage **60** between the tubing **26** and the protection tube **56**. As illustrated, the protected passage **60** may be an annular space located radially between the tubing **26** and protection tube **56**. The protected passage **60** is sized to accommodate passage/bypass of control lines **50** or other components **48**. As illustrated in FIG. 3, a given control line **50** (or other component **48**) is routed from an exterior **52**; through hole **46** of the first coupling **42**; along the interior of tubing **26**; and then out through hole **46** of the second coupling **42** so as to effectively bypass the packer **20**. If protection tube **56** is used, the given control line **50** (or other component **48**) is routed along the interior of tubing **26** in the protected passage **60** between tubing **26** and protection tube **56**.

(20) Referring generally to FIG. 4, another embodiment of packer **20** is illustrated. In this example, the tubing **26** and the tubing sections **44** have different diameters, e.g. different interior diameters. In the illustrated embodiment, tubing **26** has a larger interior diameter than that of tubing sections **44**. The couplings **42** are constructed to receive the larger diameter tubing **26** on the inner side of packer **20** and to receive the smaller diameter tubing sections **44** on the outer side of packer **20**.

(21) Depending on the parameters of a given operation and the environment in which such operation is conducted, the components of packer **20** may be made from a variety of materials and

in a variety of configurations. For example, the tubing 26 and couplings 42 may be made as a unitary component or as separate components which are connected together. Various numbers of holes/passages 46 may be constructed in various sizes and arrangements through each of the couplings 42 to accommodate the bypass functionality. The tubing 26 and corresponding tubing sections 44 of well string 22 may be formed with various diameters which are common or dissimilar. Also, the expandable metal bladder 28 and sealing element 38 may be constructed from a variety of materials and in a variety of sizes and configurations. Different types of connection techniques also may be utilized for connecting the packer components.

(22) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A system, comprising: a packer sized for placement along a well string to enable formation of a seal at a desired location in a borehole, the packer comprising: a tubing including a bore; an expandable metal bladder; a first end connection disposed about the tubing coupling a first end of the expandable metal bladder to the tubing; a second end connection disposed about the tubing and coupling a second end of the expandable metal bladder to the tubing; a sealing element mounted around the expandable metal bladder and positioned to move into sealing engagement with a surrounding borehole wall when the expandable metal bladder is sufficiently expanded via internal pressure; a first coupling partially disposed about the tubing and connected to a first end of the tubing and axially spaced from the first end connection, wherein the first coupling includes: an annular body forming a bore in fluid communication with the bore of the tubing; and a hole formed in the annular body to provide a feedthrough running from an exterior of the first coupling to the bore of the first coupling; a second coupling partially disposed about the tubing and connected to a second end of the tubing and axially spaced from the second end connection, the second coupling includes: an annular body forming a bore in fluid communication with the bore of the tubing; and a hole formed in the annular body to provide a feedthrough running from an exterior of the second coupling to the bore of the second coupling; and a protection tubular disposed inside of the tubing and coupled to an interior region of the first coupling and the second coupling.
2. The system as recited in claim 1, further comprising a bypass line routed along an interior of the tubing and out to an exterior of the tubing through the hole in the first coupling and the hole in the second coupling.
3. The system as recited in claim 2, wherein each coupling comprises a sealing connector position to form a seal between the bypass line and the coupling.
4. The system as recited in claim 2, wherein the bypass line comprises a plurality of bypass lines.
5. The system as recited in claim 2, wherein the bypass line comprises a hydraulic line.
6. The system as recited in claim 2, wherein the bypass line comprises an electric line.
7. The system as recited in claim 2, wherein the bypass line comprises a fiber optic line.
8. The system as recited in claim 2, wherein the bypass line comprises a combination of different bypass lines.
9. The system as recited in claim 1, wherein the first coupling and the second coupling are each connected to the tubing via threaded connections.
10. The system as recited in claim 1, wherein the first coupling and the second coupling are formed as a unitary part with the tubing.
11. The system as recited in claim 1, wherein the packer is connected into a well string via the first and second couplings.
12. The system as recited in claim 11, wherein the well string comprises tubing sections having the

same diameter as the tubing.

13. The system as recited in claim 11, wherein the well string comprises tubing sections having a different diameter than the tubing.

14. The system as recited in claim 1, wherein the tubing comprises an expansion port extending from an interior of the tubing to a packer chamber between the tubing and the expandable metal bladder.

15. The system as recited in claim 14, wherein flow through the expansion port is controlled by a valve.

16. The system as recited in claim 1, wherein the interior region of the first coupling includes the bore of the first coupling, and wherein the interior region of the second coupling includes the bore of the second coupling.

17. The system as recited in claim 1, wherein: the hole of the first coupling runs from the exterior of the first coupling to an annulus formed by the bore of the first coupling and the protection tubular; and the hole of the second coupling runs from the exterior of the second coupling to an annulus formed by the bore of the second coupling and the protection tubular.

18. The system as recited in claim 11, wherein the well string includes a first tubular coupled to the first coupling and a second tubular coupled to the second coupling, and wherein the hole of the first coupling provides a feedthrough from an exterior of the first tubular to the bore of the tubing and the hole of the second coupling provides a feedthrough from an exterior of the second tubular to the bore of the tubing.
